

Product Laws of Exponents

MA.7.NSO.1.1

I can use the product of powers, power of a product, power of a power and identity laws to rewrite and evaluate a numerical expression.

Last unit you took roots apart.

Today you put powers together — and the base has to match.

Warm-Up

On your own. Four minutes. No notes.

1. $\sqrt{81}$
2. Between which two whole numbers is $\sqrt{40}$?
3. Solve $3x - 7 = 11$
4. Evaluate $\left(\frac{2}{3}\right)^2$

Warm-Up

Answers.

1. $\sqrt{81}$ **9**

2. Between which two whole numbers is $\sqrt{40}$? **6 and 7**

3. Solve $3x - 7 = 11$ **$x = 6$**

4. Evaluate $\left(\frac{2}{3}\right)^2$ **$\frac{4}{9}$**

Notes

Copy the three words.

I. A power is a shorthand for repeated multiplication

A. **base** — the number used as a factor

B. **exponent** — how many times the base is a factor

C. **power** — the whole thing, base and exponent together

$$2^3 = 2 \cdot 2 \cdot 2 = 8$$

$$\left(\frac{2}{3}\right)^2 = \frac{2}{3} \cdot \frac{2}{3} = \frac{4}{9}$$

Notes

Where the rule comes from.

II. Product of powers — same base, add the exponents

$$3^2 \cdot 3^3 = (3 \cdot 3)(3 \cdot 3 \cdot 3) = 3^5$$

$$3^2 \cdot 2^3 = (3 \cdot 3)(2 \cdot 2 \cdot 2) \quad \text{— different bases, no shortcut}$$

Product of powers: $a^m \cdot a^n = a^{m+n}$

Check with values: $3^2 \cdot 3^3 = 9 \cdot 27 = 243 = 3^5$

Notes

III. Power of a product — the exponent goes to every factor

$$(2 \cdot 5)^3 = (2 \cdot 5)(2 \cdot 5)(2 \cdot 5) = (2 \cdot 2 \cdot 2)(5 \cdot 5 \cdot 5) = 2^3 \cdot 5^3$$

Power of a product: $(ab)^m = a^m \cdot b^m$

Works for a product only, never for a sum:

$$(10 + (-2))^3 = 8^3 = 512 \quad \text{but} \quad 10^3 + (-2)^3 = 992$$

Notes

IV. Power of a power, and the exponent 1

Power of a power	Repeated multiplication	Single power
$(4^2)^3$	$4^2 \cdot 4^2 \cdot 4^2$	4^6
$(10^4)^4$	$10^4 \cdot 10^4 \cdot 10^4 \cdot 10^4$	10^{16}
$(3^1)^5$	$3 \cdot 3 \cdot 3 \cdot 3 \cdot 3$	3^5

Power of a power: $(a^m)^n = a^{mn}$

Identity: $a^1 = a$

Example 1

Two ways to the same value.

$$3^2 \cdot 3^3$$

Which law? Then find the value.

Example 1

Worked both ways.

$$3^2 \cdot 3^3 = 3^{2+3} = 3^5 \quad \text{product of powers: same base, add}$$

$$3^5 = 243 \quad \text{evaluate the single power}$$

$$9 \cdot 27 = 243 \quad \text{check: evaluate each, then multiply}$$

Answer: 243

Your Turn

Same steps, your numbers. Boards up when done.

$$2^3 \cdot 2^4$$

Your Turn

Answer.

$$2^3 \cdot 2^4$$

same base, add: 2^7

Answer: $2^7 = 128$

Example 2

A negative base and a power of a power.

$$[(-2)^3]^2$$

Which law? Then find the value.

Example 2

Worked.

$$[(-2)^3]^2 = (-2)^{3 \cdot 2} = (-2)^6 \quad \text{power of a power: multiply}$$

$$(-2)^6 = 64 \quad \text{six negative factors — positive}$$

$$(-8)^2 = 64 \quad \text{check: } (-2)^3 = -8, \text{ then square}$$

Answer: 64

Your Turn

Same steps, your numbers. Boards up when done.

$$[(-3)^2]^3$$

Your Turn

Answer.

$$[(-3)^2]^3$$

multiply the exponents: $(-3)^6$

Answer: $(-3)^6 = 729$

Whiteboards • Question 1 of 9

Boards up on three.

$$5^2 \cdot 5^3$$

Answer it.

Write it as one power.

Whiteboards • Question 1 of 9

Answer.

$$5^2 \cdot 5^3$$

same base, so add the exponents

Answer: 5^5

Whiteboards • Question 2 of 9

Boards up on three.

$$2^4 \cdot 2^2$$

Answer it.

Find the value.

Whiteboards · Question 2 of 9

Answer.

$$2^4 \cdot 2^2$$
$$2^6$$

Answer: 64

Whiteboards • Question 3 of 9

Boards up on three.

$$(2 \cdot 5)^3$$

Answer it.

Find the value.

Whiteboards • Question 3 of 9

Answer.

$$(2 \cdot 5)^3$$

$$2^3 \cdot 5^3 = 8 \cdot 125, \text{ or } 10^3$$

Answer: 1000

Whiteboards • Question 4 of 9

Boards up on three.

$$(3^2)^4$$

Answer it.

Write it as one power.

Whiteboards • Question 4 of 9

Answer.

$$(3^2)^4$$

four copies of 3^2 , so eight 3s

Answer: 3^8

Whiteboards • Question 5 of 9

Boards up on three.

$$4^3 \cdot 4^2$$

Which expression is equivalent?

A. 4^5

B. 4^6

C. 16^5

D. 8^5

Answer it.

Whiteboards · Question 5 of 9

Answer.

$$4^3 \cdot 4^2$$

Which expression is equivalent?

A. 4^5

B. 4^6

C. 16^5

D. 8^5

Answer: A — 4^5

Whiteboards • Question 6 of 9

Boards up on three.

$$(2^3)^2$$

Which expression is equivalent?

A. 2^6

B. 2^5

C. 4^6

D. 2^9

Answer it.

Whiteboards • Question 6 of 9

Answer.

$$(2^3)^2$$

Which expression is equivalent?

A. 2^6

B. 2^5

C. 4^6

D. 2^9

Answer: A — 2^6

Whiteboards • Question 7 of 9

Boards up on three.

$$(-3)^2 \cdot (-3)^1$$

Answer it.

Find the value.

Whiteboards • Question 7 of 9

Answer.

$$(-3)^2 \cdot (-3)^1$$

$(-3)^3$ — three negative factors

Answer: -27

Whiteboards • Question 8 of 9

Boards up on three.

$$\left(\frac{1}{2}\right)^2 \cdot \left(\frac{1}{2}\right)^3$$

Answer it.

Find the value.

Whiteboards • Question 8 of 9

Answer.

$$\left(\frac{1}{2}\right)^2 \cdot \left(\frac{1}{2}\right)^3$$
$$\left(\frac{1}{2}\right)^5$$

Answer: $\frac{1}{32}$

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Aiyanna says she can use the power of a product law to evaluate $(10 + (-2))^3$. Is she right?

Write your answer in sentences.

Say what the law needs, and show the value.

Whiteboards • Question 9 of 9

Answer.

Aiyanna says she can use the power of a product law to evaluate $(10 + (-2))^3$. Is she right?

the parentheses hold a SUM, not a product

Answer: No — $(10 + (-2))^3 = 8^3 = 512$

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Stop at a SmartScore of 67 on each skill. Whatever you do not finish is tonight's practice.

Today's skills

Evaluate powers — SZQ

Multiplication with exponents — 5SK

Power rule — 8DA

Optional: Understanding exponents — BFA